

The unholy combination

EVERYONE'S TALKING ABOUT AI, BLOCK-CHAIN, IOT AND ROBOTICS being the pivotal technologies of the coming decade and they're absolutely right. What's even exciting is how one or more of these technologies can come together to create something that skirts the borders of our imagination. I know it's a little cheesy to throw in an Avengers reference but as we sat in the cinema halls towards the end of the movie *SPOILER* watching some of the heroes wither away into dust, I couldn't help but think of the reverse of Thanos' little action. What if objects could materialise out of thin air and take the shape and form of whatever you want? Something like the T-1000 from the Terminator movies, except it's not limited to liquid metal and could be formed out of any matter. There's a name for this idea - Utility Fog, and it was coined by Nanotechnology pioneer John Storrs Hall.

Utility Fog is an idea based on the concept of a large collection of tiny self-replicating robots that link up with each other to build whatever you want. These objects that are built could be permanent of just a temporary state for the Utility Fog. Imagine requiring a toadstool and all you had to do was ask the cabinet door and it would morph to become one. Moreover, you don't need to have a huge clumps of these tiny nanobots acting as one. Even having a simple layer of them plastered over a wall would mean that you can change the texture and colour of your walls as per your mood. It would be the perfect wallpaper. Are the walls catching too much dust? Tell the nanobots to carry the dust away and they'll perform peristalsis to do just that. Is the room too bright? Well then ask the window to change its tint and voila, your room is instantly darker by a few shades.

The Utility Fog need not be relegated to simple roles. You could wear one and ask it to interact with the world in whatever way you deem fit. Ask it to become your helmet as you mount your bike or ask it to become a make-shift bunch of roses because you've forgotten your anniversary. You can take things up a notch and ask it to stitch together a flesh would and if things are advanced and small enough, you may even ask them to enter your

bloodstream and clear artery blockages. It could be one of the most non-invasive invasive form of surgery. Also, it need not be just a temporary arrangement, we could have nanobots in our system just as part of our everyday lives. We're already married to our smartphones, so having a similar relationship with millions of nanobots coursing through our veins isn't a far-fetched concept.

The logical next step would be for these nanobots to permeate the blood-brain barrier and latch onto our brains and form an interface, a BCI (brain-computer interface). Since the nanobots can talk to each other, what we're looking at is a hive-mind system. Imagine the speed with which information would flow from person to person. That too truly. We already have rapid means to information transportation i.e. the Internet but it can only get information from one human to another. It doesn't necessarily mean that the information would be understood by the recipient. With a BCI, the information would be understood. Imagine every human being in existence instantly understanding Einstein's theory of special relativity. Imagine every human being capable of crunching numbers as fast as the nearest computer. Yeah, by the time we have nanobots in our blood, we will have computers that are faster and smarter than human beings.

There would be no need for wars, the last war to be fought would just be mass-bombing with nanobot payloads that would infect each and every one, bringing them into the fold. Into... the hive mind. The concept of individuality will be lost as we yearn to overcome our flaws. As we aim to achieve... perfection. And thus, humanity ceases to exist and a new race of interconnected beings come to be. Thus we, the borg, are born. **d**



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